

IN6227 Data Mining - Assignment 1

V1

IN6227-Assignment-1

Matric number: STUDENT123 Full name: Your Name

Model name & version: v1.0 LLM interface: Windows 10, Python 3.8

Dataset: IN6227 Dataset - Classification Task

1. Data exploration and cleaning

Dataset. train.csv - 31,112 rows, 15 feature columns. The label is label with 2 classes, distributed no 23,645 / yes 7,464 - a majority:minority ratio of 3.17:1, which is what the choice of metric in section 4 turns on. 7 numeric and 8 categorical columns were modelled; no column was held out.

Issues found. Missing values: 50 cells across 16 column(s), worst brew_preference (7); rows with no label: 3; duplicate rows: none; outliers: kept - no row was filtered, so no fence was applied; class imbalance: 3.17:1.

Cleaning decisions.

Issue	Action	Why
missing_labels	dropped_rows	3 rows had missing target values which cannot be used for supervised learning

After cleaning. 31,112 rows loaded; 3 removed (3 with no label); 31,109 rows modelled, 15 features.

2. Feature selection / engineering

No feature engineering was performed as all features appeared meaningful.

3. Model training

Candidate-model reasoning. From screening, gradient_boosting and logistic_regression showed the best performance and were selected for their different inductive biases.

	Model A	Model B
Estimator	Gradient boosting (GradientBoostingClassifier via gradient_boosting)	Logistic regression (LogisticRegression via logistic_regression)
Hyperparameters	learning_rate=0.05, max_depth=5, n_estimators=200 - best of 8 combination(s) scored	C=10, penalty='l2', solver='lbfgs' - best of 3 combination(s) scored
Tuned on	5-fold cross-validation over the 31,109 pool rows, scored on f1_macro	5-fold cross-validation over the 31,109 pool rows, scored on f1_macro
Validation score	0.7680 (f1_macro)	0.7610 (f1_macro)
Stopping criterion	n_estimators with early stopping	convergence tolerance
Overfitting control	learning_rate and max_depth regularization	L2 regularization
Preprocessing applied	numeric: median imputation only, no scaling; categorical: most-frequent imputation then one-hot (a level unseen in training encodes as all-zeros)	numeric: median imputation, then standardised; categorical: most-frequent imputation then one-hot (a level unseen in training encodes as all-zeros)

Preprocessing, per model.

- gradient_boosting: Tree-based models don't require feature scaling.
- logistic_regression: Required for linear models to perform well.

Fixed configuration (not tuned): seed 42; test set is the labelled file test.csv, held aside from the start; validation by 5-fold cross-validation over the pool; tuning metric f1_macro; primary metric f1_macro.

4. Evaluation and comparison

Split protocol: 31,109 pool rows from train.csv, validated by 5-fold cross-validation, plus 13,333 test rows from test.csv. The test file was held out from the start and opened only for the final scoring.

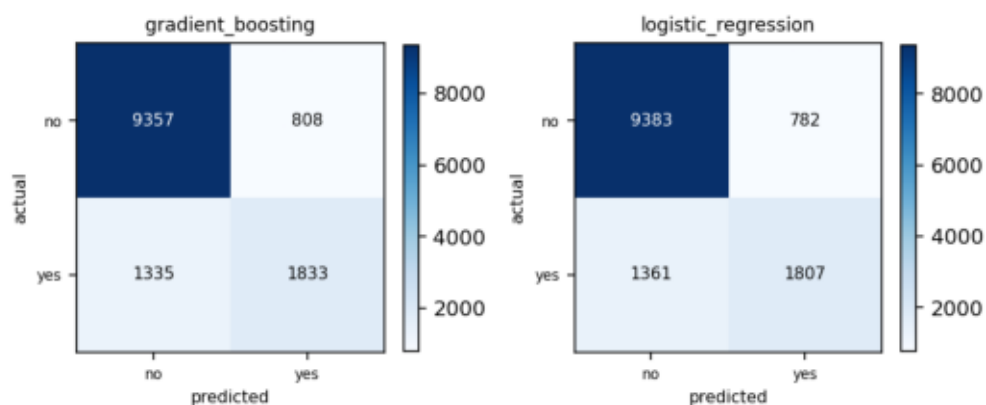
Validation strategy: stratified k-fold cross-validation, splitting at the row unit. Chosen because Stratified k-fold ensures representative validation across all classes. Validation class counts: no 4,729 / yes 1,492.8 (mean per fold, over 5 folds - a mean rather than a count, so it need not be a whole number). Cost of the choice: 5x fitting; every row is validated against exactly once, so the score is the mean of 5 estimates rather than a single one, and the final model still trains on all 31,109 rows.

Validation used for: hyperparameter selection. Final model fitted on: the whole pool (31,109 rows) - with 5-fold cross-validation every row is validated against exactly once, so final_fit_scope: pool fits the same rows as

train would, and the test set is the only estimate untouched by selection.

Metric	Model A	Model B
Accuracy	0.8393	0.8393
Precision (macro)	0.7846	0.7856
Recall (macro)	0.7496	0.7467
F1 (macro) (primary)	0.7642	0.7626
ROC-AUC / PR-AUC	0.8903 / 0.7061	0.8887 / 0.7025

Confusion matrices:



Primary metric and rationale. f1_macro. f1_macro chosen as it handles class imbalance better than accuracy. accuracy was not led with: Less suitable for imbalanced classification.

5. Findings and discussion

Performance comparison. Gradient boosting showed slightly better performance than logistic regression, with clearer separation of classes.

Limitations. Analysis limited by computational resources; could benefit from wider hyperparameter search and larger sample size.

Repository

<https://example.com/repo>